

Accuracy of The Cancer Genome Atlas Classification vs American Joint Committee on Cancer Classification for Prediction of Metastasis in Patients With Uveal Melanoma

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 Supplemental content

IMPORTANCE The Cancer Genome Atlas (TCGA) classification is a newly emerging method for prediction of uveal melanoma (UM)-related metastasis and death. Limited information is available regarding the accuracy of the TCGA classification for prediction of metastasis in patients with UM.

OBJECTIVE To investigate the accuracy of the TCGA classification for predicting UM-related metastasis compared with the American Joint Committee on Cancer (AJCC) classification.

DESIGN, SETTING, AND PARTICIPANTS In this retrospective cohort study, patients with UM treated with plaque radiotherapy at a tertiary referral center from October 1, 2008, to December 31, 2018, were evaluated. All patients with tumors classified according to the American Joint Committee on Cancer Staging Manual, 8th Edition, and who completed pretreatment fine-needle aspiration biopsy sampling for genetic analysis of chromosomes 3 and 8 for TCGA classification were included. Tumors were classified into 4 categories, 17 subcategories, and 4 stages using AJCC classification and further grouped into 4 classes using TCGA classification.

MAIN OUTCOMES AND MEASURES Value of TCGA classification vs AJCC classification for predicting UM-related metastasis.

RESULTS Of 642 included patients, 331 (51.6%) were women, and the mean (SD) age was 58.0 (13.8) years. There were 642 tumors from 642 patients classified according to both AJCC and TCGA classifications. The mean (range) follow-up time for the entire cohort was 43.7 (1.4-159.2) months. At 5 years, TCGA classification showed higher value for prediction of metastasis (4 TCGA classes: Wald statistic, 94.8; hazard ratio [HR], 2.8; 95% CI, 2.3-3.5; $P < .001$; 4 AJCC categories: Wald statistic, 67.5; HR, 2.6; 95% CI, 2.1-3.2; $P < .001$; 17 AJCC subcategories: Wald statistic, 74.3; HR, 1.3; 95% CI, 1.2-1.3; $P < .001$; 4 AJCC stages: Wald statistic, 67.0; HR, not applicable; $P < .001$). After entering TCGA and AJCC classifications into a multivariate model, TCGA classification still showed higher value for prediction of metastasis (TCGA classification: Wald statistic, 61.5; HR, 2.4; 95% CI, 1.9-2.9; $P < .001$; AJCC classification: Wald statistic, 35.5; HR, 1.9; 95% CI, 1.5-2.4; $P < .001$).

CONCLUSIONS AND RELEVANCE These results suggest that TCGA classification provides accuracy that is superior to AJCC categories, subcategories, and stages for predicting metastasis from UM. When genetic testing is available, TCGA classification can provide a more accurate way to identify patients at high risk of metastasis who might benefit from adjuvant therapy.

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Uveal melanoma (UM) is the most common primary intraocular malignant tumor in adults, with favorable local tumor control in the era of plaque radiotherapy but relatively poor systemic outcomes with moderate metastatic risk because of micrometastasis before treatment.^{1,2} This insidious behavior has spawned several clinical trials evaluating adjuvant therapy for prevention of metastasis in high-risk patients with UM.^{3,4} To best identify patients at high risk, ocular oncologists and medical oncologists have assessed various single or combined parameters as predictors of metastatic disease.⁵⁻⁷ Among the numerous systems for prediction of UM-related metastasis, some strive for maximal accuracy at the expense of simplicity, while others are easier to use at the expense of precise predictive value.⁸⁻¹⁰

The American Joint Committee on Cancer (AJCC) Tumor-Node-Metastasis (TNM) staging is currently the benchmark for categorizing patients with various types of solid tumors, including UM, in terms of predicting prognosis.¹¹ This system combines tumor size, ciliary body involvement, and extraocular extension to divide UM into 4 tumor categories, 17 tumor subcategories, and 4 stages.¹² However, rapid evolution of our understanding of cancer biology, such as identification and validation of several genetic and molecular biomarkers that can estimate patient outcomes more accurately, has caused some oncologists to question the efficacy of an AJCC-based approach in clinical care at a personalized level. This important question has been highlighted for UM, which shows unique behavior in that UM does not follow the usual stepwise anatomical progression (tumor to lymph node to distant metastasis), which is the mainstay of the AJCC staging system. Instead, UM can develop clinically undetectable micrometastasis to the liver, lung, and bone marrow at a very early course in the disease and remain in those sites as a dormant tumor with potential for late recurrence.^{2,13}

The advent of genetic analysis using fine-needle aspiration biopsy (FNAB) has changed the landscape of UM prognostication, providing genetic biomarkers as modern and valuable parameters.¹⁴⁻¹⁶ Given this new genetic information, while some investigators have tried to improve the accuracy and prognostic value of AJCC staging via incorporating genetic and/or cytologic parameters,^{17,18} other authors have tried to introduce new prognostic algorithms based solely on genetic factors.⁸⁻¹⁰ For example, 2 innovative web-based tools (Liverpool Uveal Melanoma Prognosticator Online [LUMPO]¹⁷ and Predicting Risk of Metastasis in Uveal Melanoma [PriMeUM]¹⁹) have been recently developed to provide an individualized estimation of overall survival and risk of metastasis in patients with UM. Although all of these valuable investigations brought new information to the field by improving the accuracy of detecting high-risk patients, they still experience one of the main drawbacks of the AJCC TNM classification, ie, complexity, making them difficult and time-consuming to use in an everyday clinical practice setting.

In 2005, the National Cancer Institute Center for Cancer Genomics and the National Human Genome Research Institute conceived The Cancer Genome Atlas (TCGA) project for 33 cancer types to achieve better insight into the genetic and immunologic makeup of these malignancies. In 2010, Damato

Key Points

Question How does The Cancer Genome Atlas (TCGA) classification compare with the American Joint Committee on Cancer (AJCC) classification for predicting metastasis in patients with uveal melanoma?

Findings In this cohort study of 642 patients, TCGA classification had a higher value for prediction of distant uveal melanoma metastasis compared with AJCC classification.

Meaning These results suggest that the TCGA classification is superior to the AJCC classification for robust prediction of risk for uveal melanoma-related metastasis.

et al²⁰ in an innovative study proposed detection of chromosome 1, 3, 6, and 8 abnormalities for routine clinical prognostication in UM. In 2018, according to the TCGA classification, UM (all types, including iris, ciliary body, and choroidal melanoma) was categorized into 4 classes (classes A, B, C, and D) based on the presence or absence of chromosome 3 monosomy and the presence and degree of chromosome 8q gain. The best prognostic class (class A) demonstrated disomy 3 and 8, and subsequent classes demonstrated increasing risk for metastasis, including with disomy 3 and 8q gain (class B), monosomy 3 and 8q gain (class C), and monosomy 3 and multiple 8q gains (class D).^{21,22} In 2019, we published an analysis of 658 patients with UM categorized according to TCGA guidelines based on FNAB cytogenetic results,²³ and we demonstrated that TCGA classification is a simple method for robust prediction of risk for melanoma-related metastasis and death. Herein, we investigate the accuracy of the TCGA classification as a genetic-based model compared with the AJCC classification system as a clinical-based model for UM outcomes using Cox regression models.

Methods

This study was conducted in accordance with the Declaration of Helsinki, and institutional review board approval was obtained from the Wills Eye Hospital. Written informed consent for use of data for research was obtained from all patients at first examination.

Patients

The medical and imaging records of all patients diagnosed with UM at the Ocular Oncology Service, Wills Eye Hospital, Thomas Jefferson University, Philadelphia, Pennsylvania, from October 1, 2008, to December 31, 2018, were reviewed. All patients with UM who completed FNAB for genetic analysis (analysis of chromosomes 3 and 8) were eligible. Patients without complete genetic analysis of both chromosomes 3 and 8 were excluded. Patients with iris melanoma were excluded (n = 15), as iris melanoma is addressed separately from choroidal and ciliary body melanoma by AJCC staging. All patients underwent a comprehensive ophthalmic examination by a senior ocular oncologist (C.L.S.), including slitlamp biomicroscopy and indirect ophthalmoscopy. Color fundus

photography, B-scan ultrasonography, optical coherence tomography, fundus autofluorescence, and fluorescein angiography were performed for diagnostic purposes.

AJCC Classification

Clinical tumor features included tumor anatomic location (choroid and ciliary body), anterior and posterior margin (macula, macula-equator, equator-ora serrata, and ciliary body), distance to optic disc (in millimeters), distance to foveola (in millimeters), largest basal diameter (in millimeters), and previous photographic evidence of tumor growth. Tumor thickness was measured using B-scan ultrasonography (in millimeters). Based on the tumor largest basal diameter, thickness, location, and extraocular extension, all tumors were classified into 4 tumor categories, 17 tumor subcategories, and 4 stages according to the American Joint Committee on Cancer Staging Manual, 8th Edition (eFigures 1, 2, and 3 in the [Supplement](#)).¹²

TCGA Classification

Samples for genetic analysis were obtained using FNAB, which was performed in the operating room under sterile conditions using a 10-mL syringe connected to a 27-gauge needle delivered into the melanoma via trans pars plana or trans scleral approach. Samples were stored in Hanks balanced salt solution (Gibco; Life Technologies) at 4°C and submitted to the Genetic Diagnostic Laboratory at the University of Pennsylvania, Philadelphia, for chromosomal copy number analysis. The DNA Microkit (Qiagen) was used to isolate genomic DNA from the cytologic specimen using the manufacturer's listed protocol. For each sample, analysis of chromosome 3 (disomy or monosomy) and 8 (disomy, 8q gain, or 8q gain multiple) was recorded.

Based on the genetic results, each eye was classified according to the TCGA classification for UM as classes A, B, C, or D, with the most advanced tumors in class D. Class A included melanoma with disomy of chromosomes 3 and 8; class B, disomy of chromosome 3 and chromosome 8q gain; class C, monosomy of chromosome 3 and chromosome 8q gain; and class D, monosomy of chromosome 3 and multiple chromosome 8q gains.^{21,22}

Surveillance

Each patient was advised to be monitored for systemic metastasis by a medical oncologist with physical evaluation and liver function tests twice yearly and chest radiography and magnetic resonance imaging once yearly. Melanoma-related metastasis was determined by medical oncology records and imaging, which was typically confirmed by tissue diagnosis. Cause of death was determined by a medical oncologist, primary care physician, or autopsy report.

Statistical Analysis

Statistical analysis was performed using SPSS Statistics Software version 18 (IBM). Comparison between 4 AJCC tumor categories (T1, T2, T3, and T4) and 4 AJCC stages (I, II, III, and IV) was performed using χ^2 test (and Fisher exact test when indicated) for categorical variables and 1-way analysis of variance test for continuous variables. Spearman test was used to find the correlation between TCGA and AJCC classifications. Uni-

variate Cox regression analysis was performed for each independent variable, including TCGA class (A, B, C, or D), AJCC tumor category (T1, T2, T3, or T4), AJCC T subcategory (T1a, T2a, T2b, etc), AJCC stage (stage I, II, III, or IV), largest basal diameter, thickness, ciliary body involvement, extraocular extension, age, and sex, to find the predictors of metastasis with the corresponding hazard ratios (HRs) and Wald statistics. Pairwise (within-group) HR calculation was also performed for categorical variables with more than 2 values. Each variable was subsequently entered into a multivariate Cox regression model using forward stepwise likelihood ratio method. All *P* values were 2-tailed, and a *P* value less than .05 was considered statistically significant. No adjustment to *P* values was made for multiple analyses.

Results

A total of 657 patients were enrolled in this study. There were 15 eyes with iris melanoma from 15 patients that were excluded because the AJCC classification for iris melanoma is different from that for choroidal and ciliary body melanoma. Hence, this study included 642 melanomas from 642 patients classified according to both the AJCC and TCGA classification during the study period.

Patient characteristics by AJCC tumor category and stage are listed in [Table 1](#). There were no differences among 4 tumor categories regarding age at presentation, sex, race/ethnicity, or affected eye. Presenting visual acuity was worse with more advanced tumor category. There were no trends for age and sex vs tumor categories using 1-way analysis of variance test. There were no differences among 4 AJCC stages regarding age at presentation, sex, race/ethnicity, or affected eye. Presenting visual acuity was worse with more advanced stage.

Clinical features of UM by AJCC tumor category and stage are described in [Table 1](#). Comparison showed more advanced tumor category with more anterior location, tumor epicenter less frequently in the choroid, anterior tumor margin beyond the equatorial region, and posterior margin outside the macula. More advanced tumor category demonstrated increased tumor distance to the optic disc and foveola, larger tumor basal diameter, and greater thickness. Comparison showed more advanced stage with more anterior location, tumor epicenter less frequently in the choroid, anterior tumor margin beyond the equatorial region, and posterior margin outside the macula. More advanced stage demonstrated increased tumor distance to the optic disc and foveola, larger tumor basal diameter, and greater thickness.

Clinical outcomes of patients with UM by AJCC tumor category and stage are summarized in [Table 1](#). There were 9 patients with lack of follow-up in the study for whom vital status and presence of metastasis could not be confirmed. There were 455 patients with follow-up data at 2 years, 346 with follow-up data at 3 years, and 168 with follow-up data at 5 years. The mean (range) follow-up time for the entire cohort was 43.7 (1.4-159.2) months. A comparison revealed shorter length of follow-up for patients with more advanced tumor category. More advanced tumor category showed a higher percentage

Table 1. Demographic Characteristics, Tumor Characteristics, and Outcomes of Patients With Uveal Melanoma (UM) by American Joint Committee on Cancer Staging Manual, 8th Edition, Tumor Classification

Characteristic	No. (%)		AJCC Classification								
	Tumor Category								Total (N = 642)		
	T1 (n = 247)	T2 (n = 164)	T3 (n = 177)	T4 (n = 54)	P Value	Stage I (n = 243)	Stage II (n = 314)	Stage III (n = 85)		Stage IV (n = 0)	P Value
Demographic Characteristics											
Age, y											
Mean (SD)	56.5 (13.4)	58.9 (13.5)	58.8 (13.8)	59.0 (16.4)	.22	56.5 (13.4)	58.6 (13.9)	59.8 (14.3)	NA	.10	58.0 (13.8)
Median (range)	59.0 (10.0-83.0)	61.0 (26.0-85.0)	59.0 (23.0-94.0)	61.0 (19.0-90.0)	NA	59.0 (10.0-83.0)	60.0 (23.0-94.0)	61.0 (19.0-90.0)	NA	NA	59.0 (10.0-94.0)
Sex											
Male	128 (51.8)	82 (50.0)	79 (44.6)	22 (41)	.33	128 (52.7)	150 (47.8)	33 (39)	0	.08	311 (48.4)
Female	119 (48.2)	82 (50.0)	98 (55.4)	32 (59)		115 (47.3)	164 (52.2)	52 (61)	0		331 (51.6)
Race/ethnicity											
White	239 (96.8)	161 (98.2)	167 (94.4)	53 (98)		235 (96.7)	303 (96.5)	82 (96)	0		620 (96.6)
African American	0	1 (0.6)	1 (0.6)	0		0	2 (0.6)	0	0		2 (0.3)
Asian	5 (2.0)	1 (0.6)	7 (4.0)	1 (2)	.86	5 (2.1)	7 (2.2)	2 (2)	0	.84	14 (2.2)
Hispanic	1 (0.4)	0	1 (0.6)	0		1 (0.4)	1 (0.3)	0	0		2 (0.3)
Other/unknown	2 (0.8)	1 (0.6)	1 (0.6)	0		2 (0.8)	1 (0.3)	1 (1)	0		4 (0.6)
Affected eye											
Right	126 (51.0)	88 (53.7)	93 (52.5)	28 (52)	.96	123 (50.6)	167 (53.2)	45 (53)	0	.86	335 (52.2)
Left	121 (49.0)	76 (46.3)	84 (47.5)	26 (48)		120 (49.4)	147 (46.8)	40 (47)	0		307 (47.8)
Visual acuity											
20/20-20/50	208 (84.2)	132 (80.5)	110 (62.1)	28 (52)		204 (84.0)	221 (70.4)	53 (62)	0		478 (74.5)
20/60-20/200	25 (10.1)	20 (12.2)	40 (22.6)	14 (26)	<.001	25 (10.3)	56 (17.8)	18 (21)	0	<.001	99 (15.4)
20/400-NLP	14 (5.7)	12 (7.3)	27 (15.3)	12 (22)		14 (5.8)	37 (11.8)	14 (16)	0		65 (10.1)
Tumor Characteristics											
Tumor epicenter											
Choroid	243 (98.4)	154 (93.9)	150 (84.7)	45 (83)	<.001	243 (100)	302 (96.2)	47 (55)	0	<.001	592 (92.2)
Ciliary body	4 (1.6)	10 (6.1)	27 (15.3)	9 (17)		0	12 (3.8)	38 (45)	0		50 (7.8)
Anterior margin											
Macula	30 (12.1)	1 (0.6)	1 (0.6)	0		30 (12.3)	2 (0.6)	0	0		32 (5.0)
Macula to equator	162 (65.6)	62 (37.8)	17 (9.6)	1 (2)	<.001	162 (66.7)	79 (25.2)	1 (1)	0	<.001	242 (37.7)
Equator to ora	44 (17.8)	73 (44.5)	74 (41.8)	6 (11)		44 (18.1)	146 (46.5)	7 (8)	0		197 (30.7)
Ciliary body and iris	11 (4.5)	28 (17.1)	85 (48.0)	47 (87)		7 (2.9)	87 (27.7)	77 (91)	0		171 (26.6)
Posterior margin											
Macula	179 (72.5)	91 (55.5)	79 (44.6)	31 (57)		179 (73.7)	169 (53.8)	32 (38)	0		380 (59.2)
Macula to equator	63 (25.5)	67 (40.9)	85 (48.0)	22 (41)	<.001	63 (25.9)	131 (41.7)	43 (51)	0	<.001	237 (36.9)
Equator to ora	1 (0.4)	5 (3.0)	12 (6.8)	1 (2)		1 (0.4)	9 (2.9)	9 (11)	0		19 (3.0)
Ciliary body and iris	4 (1.6)	1 (0.6)	1 (0.6)	0		0	5 (1.6)	1 (1)	0		6 (0.9)

(continued)

Table 1. Demographic Characteristics, Tumor Characteristics, and Outcomes of Patients With Uveal Melanoma (UM) by American Joint Committee on Cancer Staging Manual, 8th Edition, Tumor Classification (continued)

Characteristic	No. (%)		AJCC Classification								
	Tumor Category					Stage			Total (N = 642)		
	T1 (n = 247)	T2 (n = 164)	T3 (n = 177)	T4 (n = 54)	P Value	Stage I (n = 243)	Stage II (n = 314)	Stage III (n = 85)		Stage IV (n = 0)	P Value
Distance to optic disc, mm											
Mean (SD)	3.2 (3.2)	4.9 (4.3)	5.5 (4.5)	4.6 (4.1)	<.001	3.2 (3.2)	4.5 (4.1)	6.8 (5.1)	NA	<.001	4.4 (4.0)
Median (range)	3.0 (0-18.0)	4.0 (0-20.0)	5.0 (0-18.0)	4.0 (0-17.0)	NA	3.0 (0-18.0)	4.0 (0-20.0)	7.0 (0-18.0)	NA	NA	3.0 (0-20.0)
Distance to foveola, mm											
Mean (SD)	2.7 (3.1)	4.6 (4.3)	5.5 (4.3)	4.1 (3.7)	<.001	2.7 (3.1)	4.7 (4.1)	6.3 (4.8)	NA	<.001	4.0 (4.0)
Median (range)	2.0 (0-16.0)	3.0 (0-18.4)	4.0 (0-17.0)	4.0 (0-17.0)	NA	2.0 (0-16.0)	3.0 (0-18.4)	6.0 (0-17.0)	NA	NA	3.0 (0-18.4)
Largest basal diameter, mm											
Mean (SD)	8.0 (2.0)	12.2 (1.9)	15.8 (1.7)	19.4 (1.5)	<.001	8.0 (2.0)	13.9 (2.6)	17.9 (2.6)	NA	<.001	12.2 (4.3)
Median (range)	8.0 (3.0-12.0)	12.0 (8.0-17.0)	16.0 (9.0-18.0)	19.8 (16.0-24.0)	NA	8.0 (3.0-12.0)	14.0 (6.0-18.0)	19.0 (10.0-24.0)	NA	NA	12.0 (3.0-24.0)
Thickness, mm											
Mean (SD)	3.0 (0.9)	4.7 (1.3)	8.1 (2.2)	10.8 (3.2)	<.001	3.0 (0.9)	6.2 (2.4)	10.0 (3.0)	NA	<.001	5.5 (3.1)
Median (range)	2.8 (1.0-6.0)	4.6 (2.2-9.0)	8.1 (3.2-14.1)	11.5 (3.7-20.4)	NA	2.8 (1.0-6.0)	5.9 (2.2-14.1)	10.0 (3.7-20.4)	NA	NA	4.5 (1.0-20.4)
Outcomes											
Follow-up, mo ^a											
Mean (SD)	49.3 (30.0)	44.1 (27.2)	38.5 (28.4)	32.6 (27.3)	<.001	49.2 (30.2)	41.8 (28.6)	34.7 (24.9)	NA	<.001	43.7 (29.2)
Median (range)	44.9 (1.7-160.6)	41.4 (1.4-119.5)	33.4 (3.8-134.2)	23.4 (1.4-104.8)	NA	44.7 (1.7-160.6)	37.3 (1.4-134.2)	31.8 (1.4-104.8)	NA	NA	38.5 (1.4-160.6)
Metastasis	8 (3.2)	21 (12.8)	39 (22.0)	23 (43)	<.001	8 (3.3)	48 (15.3)	35 (41)	0	<.001	91 (14.2)
Time to metastasis, mo											
Mean (SD)	40.2 (29.6)	38.1 (27.5)	25.7 (28.6)	22.0 (26.7)	.01	40.2 (29.7)	30.8 (28.8)	23.4 (24.8)	NA	.04	28.8 (29.3)
Median (range)	38.6 (19.0-59.0)	39.9 (2.0-81.0)	20.4 (1.0-63.0)	17.6 (1.0-79.0)	NA	38.6 (19.0-59.0)	28.3 (2.0-81.0)	18.0 (1.0-79.0)	NA	NA	24.7 (1.0-81.0)
Death	2 (0.8)	3 (1.8)	8 (4.5)	5 (9)	.002	2 (0.8)	7 (2.2)	9 (11)	0	<.001	18 (2.8)
Time to death, mo											
Mean (SD)	33.3 (2.3)	28.9 (17.1)	48.3 (22.6)	24.4 (21.6)	.24	33.3 (2.3)	45.5 (28.4)	30.8 (17.2)	NA	.42	36.8 (21.8)
Median (range)	33.3 (32.0-35.0)	28.4 (12.0-46.0)	40.9 (31.0-102.0)	20.2 (3.0-59.0)	NA	33.3 (32.0-35.0)	46.3 (12.0-102.0)	35.0 (3.0-59.0)	NA	NA	34.9 (3.0-102.0)
Death from UM metastasis	0	3 (1.8)	5 (2.8)	4 (7)	.002	0	5 (1.6)	7 (8)	0	<.001	12 (1.9)
Time to death, mo											
Mean (SD)	NA	28.9 (17.1)	43.8 (5.6)	29.8 (20.1)	.30	NA	37.2 (16.6)	34.2 (15.7)	NA	.76	35.4 (15.4)
Median (range)	NA	28.4 (12.0-46.0)	43.4 (38.0-50.0)	23.7 (12.0-59.0)	NA	NA	46.3 (12.0-50.0)	38.2 (12.0-59.0)	NA	NA	38.3 (12.0-59.0)

Abbreviations: AJCC, American Joint Committee on Cancer; NA, not applicable; NLP, no light perception.

^a Refers to actual observed follow-up time.

Table 2. Comparison of The Cancer Genome Atlas (TCGA) Classification vs American Joint Committee on Cancer (AJCC) Classification

TCGA Classification ^a	No. (%)												
	AJCC Classification												
	Tumor Category				Spearman Correlation Coefficient	P Value	Stage				Spearman Correlation Coefficient	P Value	Total (N = 642)
T1 (n = 247)	T2 (n = 164)	T3 (n = 178)	T4 (n = 54)	Stage I (n = 243)			Stage II (n = 314)	Stage III (n = 85)	Stage IV (n = 0)				
A	187 (75.7)	87 (53.0)	48 (27.1)	10 (19)	0.46	<.001	184 (75.7)	134 (42.7)	14 (16)	0	0.43	<.001	332 (51.7)
B	27 (10.9)	22 (13.4)	30 (16.9)	9 (17)			26 (10.7)	49 (15.6)	13 (15)	0			88 (13.7)
C	22 (8.9)	34 (20.7)	46 (26.0)	15 (28)			22 (9.1)	66 (21.0)	29 (34)	0			117 (18.2)
D	11 (4.5)	21 (12.8)	53 (29.9)	20 (37)			11 (4.5)	65 (20.7)	29 (34)	0			105 (16.4)

^a Class A included melanoma with disomy of chromosomes 3 and 8; class B, disomy of chromosome 3 and chromosome 8q gain; class C, monosomy of chromosome 3 and chromosome 8q gain; and class D, monosomy of chromosome 3 and multiple chromosome 8q gains.

of metastasis and death. The mean time to metastasis was shorter with more advanced tumor category. Comparison also revealed shorter length of follow-up for patients with more advanced stage. More advanced stage showed a higher percentage of metastasis and death. The mean time to metastasis was shorter with more advanced stage.

Comparisons of AJCC tumor category and AJCC stage with TCGA classification are summarized in Table 2. In general, more advanced TCGA class was associated with more advanced AJCC tumor category and AJCC stage. The Cancer Genome Atlas classification was correlated with AJCC tumor category (Spearman correlation coefficient, 0.46; $P < .001$) and AJCC tumor stage (Spearman correlation coefficient, 0.43; $P < .001$) classifications (Table 2).

To determine the variables for prediction of metastasis, we separately entered all of the potential predictors, including TCGA class, AJCC tumor category, AJCC tumor subcategory, AJCC stage, age, sex, tumor largest basal diameter, tumor thickness, tumor location, and extraocular extension, into univariate Cox regression models (Table 3). At 5 years, TCGA classification showed superior value for prediction of distant metastasis (4 TCGA classes: Wald statistic, 94.8; HR, 2.8; 95% CI, 2.3-3.5; $P < .001$; 4 AJCC categories: Wald statistic, 67.5; HR, 2.6; 95% CI, 2.1-3.2; $P < .001$; 17 AJCC subcategories: Wald statistic, 74.3; HR, 1.3; 95% CI, 1.2-1.3; $P < .001$; 4 AJCC stages: Wald statistic, 67.0; HR, not applicable; $P < .001$). Among the other variables, tumor thickness (Wald statistic, 77.4; HR, 1.3; 95% CI, 1.2-1.3; $P < .001$), tumor largest basal diameter (Wald statistic, 73.1; HR, 1.3; 95% CI, 1.2-1.4; $P < .001$), and ciliary body involvement (Wald statistic, 14.8; HR, 2.8; 95% CI, 1.7-4.8; $P < .001$) showed value for prediction of distant metastasis but were less powerful predictors than classification by TCGA (Table 3). Age, sex, and extraocular extension did not show value for prediction of metastasis.

Subsequently, we entered TCGA classification followed by AJCC tumor subcategory classification as main predictors of metastasis with the highest Wald statistics into multivariate Cox regression model (model 1) using a forward stepwise approach (Table 3). At 5 years, TCGA classification still showed superior value for prediction of metastasis (TCGA classification: Wald statistic, 61.5; HR, 2.4; 95% CI, 1.9-2.9; $P < .001$; AJCC tumor subcategory classification: Wald statistic, 35.5; HR, 1.9; 95% CI, 1.5-2.4; $P < .001$). An additional multivariate Cox regression model (model 2) was built by

entering TCGA classification followed by the individual components from the AJCC classification system, including tumor thickness, largest basal diameter, and ciliary body involvement. The Cancer Genome Atlas classification showed superior value for prediction of metastasis (TCGA classification: Wald statistic, 55.5; HR, 2.3; 95% CI, 1.9-2.9; $P < .001$; tumor thickness: Wald statistic, 5.0; HR, 1.1; 95% CI, 1.0-1.2; $P = .03$; tumor largest basal diameter: Wald statistic, 4.8; HR, 1.1; 95% CI, 1.0-1.2; $P = .03$). Ciliary body involvement was excluded from the model owing to lack of significance on multivariate analysis (Table 3).

Discussion

In this report, we compared the value of the TCGA classification with the AJCC TNM classification for predicting metastasis using a Cox regression model and showed that TCGA classification is superior to AJCC classification. In addition, we compared the value of TCGA classification with other previously reported single clinical features of UM, including tumor thickness, largest basal diameter, ciliary body involvement, and extraocular extension, and found that TCGA classification had superior value for predicting UM metastasis.

The Cancer Genome Atlas project, which was developed to better understand the genetic and immunologic makeup of 33 types of cancers through analyzing various types of RNA, defined 4 distinct classes of UM (classes A, B, C, and D).²¹ Jager et al²² appropriately defined the 4 classes based on TCGA data. First, 2 main classes were identified by the presence or absence of 2 chromosomes 3 (disomy 3 [class A and B] or monosomy 3 [class C and D]). Each of these classes was further divided into 2 subsets with their characteristic genomic profiles based on chromosome 8 status (class A with normal chromosome 8; class B and C with chromosome 8q gain; and class D with multiple chromosome 8q gains).^{21,22} In our previous report,²³ we showed that TCGA classification is a simple method for prognostication of patients with UM. In this report, we compared the accuracy of TCGA classification with AJCC TNM staging using Cox proportional HR regression analysis and showed that TCGA was superior to AJCC in terms of predicating the risk of metastasis. While findings from our previous report fulfilled

Table 3. Cox Regression Analysis of Clinical and Genetic Prognostic Variables for Metastasis in Patients With Uveal Melanoma by The Cancer Genome Atlas (TCGA) Classification vs American Joint Committee on Cancer (AJCC) Classification

Covariate	Wald Statistic	Hazard Ratio (95% CI)	P Value
Univariate Analysis			
TCGA class	94.8	2.8 (2.3-3.5)	<.001
B vs A	7.7	3.6 (1.5-8.9)	.005
C vs A	35.0	8.9 (4.3-18.2)	<.001
D vs A	81.8	24.5 (12.2-48.9)	<.001
AJCC tumor category	67.5	2.6 (2.1-3.2)	<.001
T2 vs T1	12.4	4.3 (1.9-9.8)	<.001
T3 vs T1	32.9	9.3 (4.3-19.9)	<.001
T4 vs T1	57.3	22.4 (10.0-50.2)	<.001
AJCC tumor subcategory	74.3	1.3 (1.2-1.3)	<.001
AJCC stage	67.0	NA ^a	<.001
II vs I	20.2	5.6 (2.6-11.8)	<.001
III vs I	56.5	19.1 (8.9-41.3)	<.001
IV vs I	NA ^a	NA ^a	NA ^a
Tumor characteristic			
Thickness	77.4	1.3 (1.2-1.3)	<.001
Largest basal diameter	73.1	1.3 (1.2-1.4)	<.001
Ciliary body involvement	14.8	2.8 (1.7-4.8)	<.001
Extraocular extension	0.01	1.1 (0.2-7.8)	.94
Age	2.4	1.0 (1.0-1.0)	.12
Sex	1.4	0.8 (0.5-1.2)	.24
Multivariate Analysis Model 1^b			
TCGA class	61.5	2.4 (1.9-2.9)	<.001
AJCC tumor subcategory	35.5	1.9 (1.5-2.4)	<.001
Multivariate Analysis Model 2^c			
TCGA class	55.5	2.3 (1.9-2.9)	<.001
B vs A	3.6	2.4 (1.0-6.0)	.06
C vs A	21.9	5.8 (2.8-12.2)	<.001
D vs A	47.0	13.0 (6.2-27.0)	<.001
Tumor characteristic			
Thickness	5.0	1.1 (1.0-1.2)	.03
Largest basal diameter	4.8	1.1 (1.0-1.2)	.03

Abbreviation: NA, not applicable.

^a Not calculated because stage IV indicates the presence of metastasis.

^b In model 1, AJCC tumor subcategory classification was entered as the most powerful AJCC classification predictor of metastasis along with TCGA classification in a multivariate model.

^c In model 2, tumor thickness and tumor largest basal diameter were entered as the most powerful clinical predictors of metastasis along with TCGA classification in a multivariate model.

the first important criteria of a useful predictor, ie, simplicity, our results from the present study have successfully fulfilled the second criteria of a predictor, ie, accuracy.

In this study, when we entered the variables in separate univariate Cox regression models, we found that the value (using Wald statistics) for tumor thickness (Wald statistic, 77.4; HR, 1.3; 95% CI, 1.2-1.3; $P < .001$) and tumor largest basal diameter (Wald statistic, 73.1; HR, 1.3; 95% CI, 1.2-1.4; $P < .001$) were almost the same as the value of the AJCC tumor subcategory classification (Wald statistic, 74.3; HR, 1.3; 95% CI, 1.2-1.3; $P < .001$) for prediction of metastasis. This finding is in line with previous studies, which have

found tumor thickness and tumor largest basal diameter as the most important clinical predictors of UM.^{24,25} Coupland et al²⁶ in a study of 847 patients with UM found that the correlation of extraocular extension with increased mortality was confounded by increased tumor malignancy and, in the case of posterior tumors, by more advanced disease. Hence, almost all of the predictive power of the AJCC classification system can be explained by tumor thickness and tumor basal diameter, while the other parameters used by the AJCC system (ciliary body involvement and extraocular extension) are redundant or confounded by thickness and basal diameter.

After entering each of the above-mentioned variables along with TCGA classification in a multivariate Cox regression model of time to UM metastasis together with tumor size parameters, TCGA classification had greater prognostic ability for prediction of metastasis than either measure of tumor size. In a study on 1059 patients, Shields et al⁸ found a correlation of cytogenetic profile with tumor thickness and showed a strong correlation between thickness and chromosome alterations. They demonstrated that increasing melanoma size would lead to greater cytogenetic alterations. Interestingly, it has been proven that the converse can also be true; Damato and Coupland²⁷ in a study on 1776 patients with UM demonstrated that early cytogenetic alterations can lead to more rapid tumor growth and, hence, increasing tumor size. Taken as a whole, multivariate analysis demonstrated that TCGA classification was superior to AJCC classification for prediction of time to metastasis from UM.

Limitations

We acknowledge limitations inherent in our study, including its retrospective nature and small number of outcome events (metastasis and death). In fact, given only 19 total deaths, with unknown cause in 6 cases, we could not perform a reliable regression analysis to determine the predictability of TCGA vs AJCC classification for overall or disease-specific mortality. In addition, given that no patient presented with AJCC stage IV disease (metastasis present) at date first seen, we could not calculate the HR for stage IV compared with less advanced stages. Another limitation is the short potential follow-up time for patients diagnosed with UM relatively recently. Only 168 patients were eligible for 5 or more years of follow-up by the cutoff date for this analysis.

Conclusions

In conclusion, we compared the accuracy of the newly emerged TCGA method of classification with the AJCC TNM staging system for predicting metastasis from UM, and our findings suggest that TCGA classification is a simple, more accurate model for prediction of metastasis in patients with UM. When genetic testing results are available, the TCGA classification system might be a more accurate way to identify patients at high risk of metastasis who might benefit from adjuvant therapy.

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